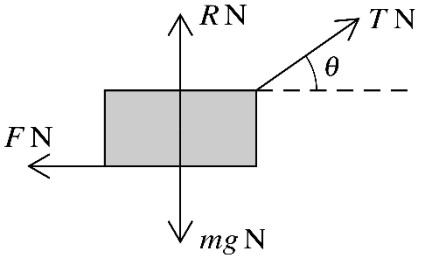
| Question | | | Answer | Marks | AO | Guidance |
|---|---|---|---|---|---|---|
| 16 | (a) | |  | | | |
| | | | Resolve vertically $R = mg - T\sin\theta$ | **B1** | **3.1a** | Must be explicit – may be seen on the diagram.<br>Allow $R + T\sin\theta = mg$ if $F = \mu(mg - T\sin\theta)$ also seen |
| | | | Motion so $F = \mu R$<br>$F = \mu(mg - T\sin\theta)$ | **M1** | **3.3** | Allow only if $R$ seen explicitly or correct vertical equation seen<br>Allow $F = \mu mg$ only if $R = mg$ seen explicitly or on the diagram |
| | | | Resolve horizontally $T\cos\theta - F = ma$<br>$ma = T\cos\theta - \mu mg + T\mu\sin\theta$ | **M1** | **3.1a** | All forces correct and no extras. Allow sign errors |
| | | | $a = \frac{T}{m}\cos\theta - \mu g + \frac{T}{m}\mu\sin\theta$ | **A1** | **2.1** | **AG** Complete argument needed |
| | | | | **[4]** | | |

| Question | | | Answer | Marks | AO | Guidance |
|---|---|---|---|---|---|---|
| 16 | (b) | | $\frac{da}{d\theta} = -\frac{T}{m}\sin\theta + \frac{T}{m}\mu\cos\theta$ | M1 | 3.1a | Attempt to differentiate wrt $\theta$ |
| | | | $-\frac{T}{m}\sin\alpha + \frac{T}{m}\mu\cos\alpha = 0$ | M1 | 1.1a | Equate their derivative to 0 and attempt to rearrange using a trig identity.<br>Condone using $\theta$ not $\alpha$ |
| | | | $\frac{\sin\alpha}{\cos\alpha} = \mu$ so $\alpha = \tan^{-1}\mu$ | A1 | 1.1 | Must be $\alpha =$<br>Also allow for $\alpha = \frac{\pi}{2} - \tan^{-1}\frac{1}{\mu}$ |
| | | | **Alternative solution** | | | |
| | | | Maximum $a$ when $\frac{T}{m}(\cos\theta + \mu\sin\theta)$ is max | | | |
| | | | Acceleration is $(R\cos(\theta - \beta))$ where $\beta = \tan^{-1}\mu$ | M1<br>M1 | | Uses trig identity<br>Attempt to find the value of $\beta$ |
| | | | Max acceleration when $(\alpha - \beta) = 0$ | | | |
| | | | Giving $\alpha = \tan^{-1}\mu$ | A1 | | Must be $\alpha =$<br>Also allow for $\alpha = \frac{\pi}{2} - \tan^{-1}\frac{1}{\mu}$ |
| | | | | [3] | | |